Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-I-Q03] | Part A (Q1-Q5)

Answer: 011

1. Equal binomial coefficients cancel in $\binom{8}{3}p^3(1-p)^5 = \binom{8}{5}p^5(1-p)^3/25$, leaving $(1-p)^2 = p^2/25$.

2. Positivity gives $1-p = p/5$, so $p = 5/6$ and the answer is 11.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2010-I-Q04] | Part A (Q1-Q5)

Answer: 515

1. The generating polynomial for one head count is $[(1+t)/2]^2(3+4t)/7 = (3+10t+11t^2+4t^3)/28$.

2. Independence makes the matching probability the sum of the squared coefficients: $(3^2+10^2+11^2+4^2)/28^2 = 123/392$.

3. The requested sum is 515.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2006-II-Q10] | Part B (Q6-Q10)

Answer: 831

1. Each team has five other games, giving $2^{10} = 1024$ equally likely joint outcomes. Let remaining-win counts be X, Y; A finishes ahead iff $X \geq Y$.

2. The equality count is $\sum_{j=0}^{5} \binom{5}{j}^2 = \binom{10}{5} = 252$. Symmetry splits the remaining outcomes evenly. Thus probability $(1024 + 252)/(2 \cdot 1024) = 319/512$, yielding 831.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2008-I-Q09] | Part B (Q6-Q10)

Answer: 190

1. If counts of the three heights are u, v, w, then $u+v+w=10$ and $3u+4v+6w=41$. Thus $v+3w=11$, giving $(u,v,w)=(1,8,1),(3,5,2),(5,2,3)$.

2. Multinomial counts are $90, 2520, 2520$, totaling 5130. Divide by 3^{10} and reduce: probability $190/3^7$, so $m=190$.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-II-Q08] | Part B (Q6-Q10)

Answer: 041

1. First-six probability at roll k is $p_k = (5/6)^{k-1}/6$. Equal counts have probability $\sum p_k^2 = (1/36)/(1 - 25/36) = 1/11$.

2. One specified player needing exactly one more roll contributes $(5/6)\sum p_k^2 = 5/66$. Both directions give total $1/11 + 10/66 = 8/33$, so the answer is 41.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2021-II-Q08] | Part B (Q6-Q10)

Answer: 049

1. All $3 \cdot 2^7 = 384$ allowed paths are equally likely. Count paths by states (B_h, B_v, T_h, T_v), recording bottom/top face and horizontal/vertical last edge.

2. The update is $(B_h + 2B_v, T_h, T_h + 2T_v, B_h)$, because a vertical edge must be followed horizontally, whereas a horizontal edge allows one horizontal and one vertical continuation.

3. Starting after one step at $(2, 0, 0, 1)$, successive states through steps 2 to 8 are $(2, 0, 2, 2)$, $(2, 2, 6, 2)$, $(6, 6, 10, 2)$, $(18, 10, 14, 6)$, $(38, 14, 26, 18)$, $(66, 26, 62, 38)$, $(118, 62, 138, 66)$.

4. The top count is 204, so the probability is $204/384 = 17/32$, giving 49.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2001-II-Q11] | Part C (Q11-Q15)

Answer: 341

1. Win-loss reversal is a symmetry, so the desired probability is half the probability of unequal counts.

2. For equal counts k wins and k losses, $k = 0, 1, 2, 3$, the sequence counts are $6!/[k!^2(6-2k)!] = 1, 30, 90, 20$, totaling 141.

3. Thus the probability is $(729 - 141)/(2 \cdot 729) = 98/243$, and the sum is 341.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2002-II-Q12] | Part C (Q11-Q15)

Answer: 660

1. Let $D(n, j)$ count admissible prefixes of length n with j hits. Set $D(0, 0) = 1$, update by a miss or hit, and retain only $j \leq \lfloor 2n/5 \rfloor$ for $n < 10$.

2. The counts at maximum permitted hit totals for $n = 3, 4, 5, 6, 7, 8, 9$ are $1, 2, 2, 5, 9, 9, 23$. The tenth shot must hit, giving 23 admissible four-hit sequences.

3. Each has probability $(2/5)^4(3/5)^6$. Thus the total is $2^4 3^6 \cdot 23/5^{10}$ and the requested value is $(2 + 3 + 23 + 5)(4 + 6 + 10) = 660$.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2003-II-Q13] | Part C (Q11-Q15)

Answer: 683

1. Let p_j be the probability of being at the start after j steps. Then $p_0 = 1$ and $p_{j+1} = (1 - p_j)/2$.

2. The fixed point is $1/3$, so $p_j - 1/3 = (2/3)(-1/2)^j$. Therefore $p_{10} = 171/512$, giving 683.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2014-I-Q11] | Part C (Q11-Q15)

Answer: 391

1. Use $u = x + y, v = x - y$. The four moves realize each pair of signs $(\Delta u, \Delta v) \in \{\pm 1\}^2$ with probability one quarter. Thus these coordinates are independent six-step sign walks.

2. The target is $u = 0$ or $v = 0$. Each event has probability $\binom{6}{3}/2^6 = 5/16$, and their intersection has the square of this probability.

3. Inclusion-exclusion gives $2(5/16) - (5/16)^2 = 135/256$, whose parameter sum is 391.

Source / 来源